
Scientific Research Experience

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| Sept 2025–Present | Canadian Institute for Theoretical Astrophysics
Analyzing JWST mid-infrared spectral data of Galactic Center Wolf-Rayet stars and their stellar winds feeding Sagittarius A*; first systematic mid-IR IFS view of the Galactic Center. Publication expected 2027. |
| May 2025–Sept 2025 | Institute for Quantum Computing
Contributed to mathematical model of polarization-dependent Mie/Rayleigh-Gans scattering in the retinal nerve fiber layer; potential proxy for early glaucoma detection. Research supported by Undergraduate Research Award. |
| Sept 2024–June 2025 | Quantum Condensed Matter Physics Group (UToronto)
Theoretically examined chiral, collapsed, single-walled carbon nanotubes as quasi-1D electronic systems with reduced moiré disorder. |
| May 2024–Sept 2024 | Winnipeg Institute for Theoretical Physics
Investigated role of light spectator scalar fields in amplifying primordial curvature perturbations, enabling primordial black hole formation as a dark matter candidate. arXiv preprint. |
| Aug 2020–March 2023 | Nuclear and Particle Physics (UWinnipeg)
Theoretically examined probabilities associated with fast neutron absorption in various nuclides and alternative nuclear poisons. Research performed through internship. |
| Sept 2020–Dec 2022 | Inorganic Materials Chemistry (UManitoba)
Identified salts for high-efficiency hydrogen production via water-splitting electrolysis; developed structure-property model to predict salt relevance. |

Research Output

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| 2026 | Ireland, Z. “First Mid-Infrared Characterization of the Galactic Center Wolf-Rayet Star Population.” <i>Canadian Association of Physicists Congress</i> , University of Ottawa, 2026. Contributed oral presentation. |
| 2026 | Ireland, Z. “First Mid-Infrared Characterization of the Galactic Center Wolf-Rayet Star Population.” <i>Physics Undergraduate Conference (PhUnC)</i> , Western University, 2026. Oral presentation. |
| 2025 | Lorenzoni, D. L., S. R. Geller, Z. Ireland, D. I. Kaiser, E. McDonough, and K. A. Wittmeier. “Light Scalar Fields Foster Production of Primordial Black Holes.” <i>arXiv preprint</i> (2025). Under review at <i>Physical Review Letters</i> . https://doi.org/10.48550/arXiv.2504.13251 |
| 2025 | Ireland, Z. “Collapsed, Chiral, Single-Walled Carbon Nanotubes as Quasi-1D Electronic Systems.” <i>Theory Canada 17</i> , University of Regina, 2025. Contributed oral presentation. |
| 2025 | Ireland, Z. “Collapsed, Chiral, Single-Walled Carbon Nanotubes as Quasi-1D Electronic Systems.” <i>Canadian Association of Physicists Congress</i> , University of Saskatchewan, 2025. Contributed oral presentation. |
| 2024 | Ireland, Z., and K. A. Wittmeier. “Light Scalar Fields Foster Production of Primordial Black Holes.” <i>Winnipeg Institute for Theoretical Physics</i> , 2024. Oral presentation. First Place Oral Presentation Award. |
| 2023 | Ireland, Z. “Quantifying the Likelihood of Neutron Radiation Absorption Utilizing Radiative Capture Cross-Sections and Number Density.” <i>CSFJ</i> (2023). |

Proficiencies

Programming Languages: Python (notably Astropy, Specutils, SciPy), LaTeX, Shell Scripting

Software & Tools: QFitsView, GitHub, VS Code, Jupyter, Interspec, Mathcha, Excel

Certifications: WHMIS, TDG, cleanroom, contamination control, laser handling, forklift, biological safety, SAFE Work, propane handling/filling

Education

University of Toronto *2022-Present*
B.Sc. (Hons) expected 2027. Physics Specialist; Minor in Mathematics.

Supplementary Education

Institute for Quantum Computing *2025*
Undergraduate School on Experimental Quantum Information Processing

McDonald Canadian Astroparticle Physics Research Institute *2025*
Canadian Astroparticle Physics Summer School

Munk School of Global Affairs & Public Policy (UToronto) *2022-2023*
Global Affairs and Public Policy Foundations (Ones) Program

Lassonde School of Engineering *2021*
Bergeron Entrepreneurs in Science and Technology program

Perimeter Institute *2021*
International Summer School for Young Physicists (full-ride)

Recent Honours

- 2026 **Sigma Xi Associate Membership**, The Scientific Research Honor Society
- 2026 **Undergraduate Research Fellowship**, Canadian Institute for Theoretical Astrophysics
- 2025, **Samuel Beatty Physics Conference Travel Award (x2)**, University of Toronto
- 2025 **Undergraduate Research Award**, Institute for Quantum Computing
- 2024 **First Place Oral Presentation Award**, Winnipeg Institute for Theoretical Physics
- 2023 **Duke of Edinburgh's International Silver Award** (52 hours of community service, 26 hours of leadership)
- 2022 **CAP Physics Prize**
- 2022 **Governor General's Academic Medal**
- 2022 **Society of Petroleum Engineers Award for Engineering Excellence in Energy**, Canadian Educational Fdn.
- 2022 **Clifton Graham Roberts Award**, Victoria College at the University of Toronto
- 2022 **Seven Oaks Education Foundation Scholarship**
- 2021, **Provincial Gold & National Silver Medallist 2022 (x2)**, Canada-Wide Science Fair
- 2021 **Full-Ride Scholarship**, Perimeter Institute's International Summer School for Young Physicists
- 2021 **Gold CREST Award**, British Science Association

Applied Scientific or Technical Appointments

- Dec 2024-April 2026 **Associate Editor for Sciences of [Arbor Journal \(Volume 7\)](#) and TRUNK Journal (Volume 1) at the University of Toronto**
Annual publications for exceptional UofT research; selected work for publication, copyedited and peer-reviewed.
- Oct 2023-April 2026 **President/Vice of The Onion Club**
President then Vice of UofT's oldest social club, circa 1876. Organized bi-weekly debates on scientific/moral philosophy, off-campus excursions, and annual campaigns for jack.org—a charity supporting youth mental health. Elected by members; oversaw 58 people.
- Sept 2020-April 2025 **Registered Tutor for Physics and Mathematics**
Tutored grades 6-11 independently and through employment with Seven Oaks Immigrant Services; established as a registered tutor at the University of Toronto, completed training (~4 months) in education and teaching.
- May 2023-Aug 2023; May 2024-Aug 2024 **Innovair Industrial Ltd. Gas Filling Plant**
Mixed/filled gases and cryogenic liquids (Ar, O₂, N₂, CO₂, He, C₃H₈) for industrial use; conducted laboratory tests to verify purity, proper blending/composition, and compliance with hydrocarbon specifications.
- Sept 2017-Dec 2022 **Royal Canadian Sea Cadets, 49 John Travers Cornwell VC**
Petty Officer 1st Class; Drill Team, Marksmanship Team, and Guard (a ceremonial unit). Facilitated city-wide parades, volunteered, taught seamanship classes. Obtained highest qualifications in marksmanship; completed Basic Training and Basic Seamanship at HMCS Quadra.
- Sept 2020-June 2021 **Winnipeg Prosthetics and Orthotics (WinPO)**
Collaborated with clinicians to develop pamphlet explaining 3-point-pressure systems in orthotics; pamphlet distributed at clinic.